

Naiyu Yin

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SUMMARY

Machine learning researcher focused on *causal discovery and inference*, *mechanistic interpretability of large language models (LLMs)* via circuit learning and sparse autoencoders, alongside *causal representation learning*, *Bayesian inference and uncertainty quantification*, and *foundation models for science*. First-author publications at **CVPR**, **AAAI**, **ECCV**, **EMNLP**, and **ICPR**, plus **ICML**, **IJCAI**, and **NeurIPS/ICML** workshops. Experienced in translating research into deployable methods through industry collaborations with IBM Research.

RESEARCH INTERESTS

- Foundation model pre-training and fine-tuning for scientific applications.
- Mechanistic interpretability (circuit learning and sparse autoencoders) of large language models.
- Causality-related research: causal discovery, causal inference, causal representation learning.
- Bayesian-related research: Bayesian inference, Bayesian neural networks, uncertainty quantification.

EDUCATION

Rensselaer Polytechnic Institute

Troy, USA

Ph.D. in Electrical, Computer, and Systems Engineering GPA: 3.77/4.00

May 2019 – Dec 2024

M.S. in Applied Mathematics

May 2019 – Dec 2023

Duke University

Durham, USA

M.S. in Electrical and Computer Engineering GPA: 3.85/4.00

Aug 2017 – May 2019

Beihang University

Beijing, China

B.Eng. in Electronic Information Engineering

Sep 2013 – Jun 2017

SELECTED PUBLICATIONS

- **Yin, N.**, Wang, H., Yu, Y., Gao, T., Dhurandhar, A., Lee, C.-H., & Ji, Q. (2026). CGU-Bayes: Causal graph uncertainty-guided Bayesian inference for domain generalization. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- **Yin, N.**, Wei, D., Gao, T., Dhurandhar, A., Ramamurthy, K., & Yu, Y. (2026). Scalable circuit learning for interpreting large language models. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- **Yin, N.**, Wang, H., Yu, Y., Gao, T., Dhurandhar, A., & Ji, Q. (2024). Integrating Markov blanket discovery into causal representation learning for domain generalization. *European Conference on Computer Vision (ECCV)*.
- **Yin, N.**, Gao, T., Yu, Y., & Ji, Q. (2024). Effective causal discovery under identifiable heteroscedastic noise model. *AAAI Conference on Artificial Intelligence*.
- **Yin, N.**, Yu, Y., Gao, T., & Ji, Q. (2024). Efficient nonlinear DAG learning under projection framework. *International Conference on Pattern Recognition (ICPR)*. (Oral)
- **Yin, N.**, & Yu, Y. (2026). Rational sparse autoencoder. *Mechanistic Interpretability Workshop, ICML*.
- **Yin, N.**, Liu, N., Lattimer, B. Y., Lua, J., & Yu, Y. (2025). Fake it till you make it: Multi-physics synthesis breaks the data barrier in chemical language models. *Machine Learning and the Physical Sciences Workshop, NeurIPS*.
- Cui, Z., **Yin, N.**, Wang, Y., & Ji, Q. (2022). Bayesian approaches for robust constraint-based causal discovery under insufficient data. *International Joint Conference on Artificial Intelligence (IJCAI)*.
- Yu, Y., Gao, T., **Yin, N.**, & Ji, Q. (2021). DAGs with no curl: An efficient DAG structure learning approach. *International Conference on Machine Learning (ICML)*.

INVITED TALKS

- **Yin, N.**, & Yu, Y. “Attention-Mechanism-Based Foundation Model for Causal Graph Learning.” Invited talk, *18th U.S. National Congress on Computational Mechanics (USNCCM18)* and *SIAM Annual Meeting*, 2025.

SKILLS

Programming Languages: Python (proficient), MATLAB (proficient), R, C/C++, Java.

ML / AI Frameworks and Tools: PyTorch, TensorFlow, JAX, Hugging Face, Scikit-learn, Git, Docker.

Specialized AI / ML Techniques: Causal discovery and inference, causal representation learning, Bayesian inference, Bayesian neural networks, uncertainty quantification, pre-training and fine-tuning transformer-based foundation models, mechanistic interpretability of language models, and nonlocal neural operators.

RESEARCH EXPERIENCE

Lehigh University

Bethlehem, USA

Postdoctoral Research Associate

Feb 2025 – Present, Full-time

- Built an attention-based foundation model for causal discovery that captures the mapping from observational data to causal graphs via an attention kernel, enabling efficient zero-shot causal graph inference; achieved substantial accuracy and efficiency gains over SOTA baselines on multiple benchmark datasets. (*Paper under review.*)
- Developed the *Rational Sparse Autoencoder (RSAE)*, replacing fixed SAE encoder activations with a trainable rational function to improve reconstruction and downstream fidelity for LLM interpretability. Published at the *Mechanistic Interpretability Workshop, ICML 2026*.
- Developed the *Poly4mer*, a physics-augmented chemical language model for polymer property prediction and targeted discovery. Published at the *Machine Learning and the Physical Sciences Workshop, NeurIPS 2025*.

IBM-Rensselaer Future of Computing Research Collaboration

Yorktown Heights, USA

Research Scientist Intern

May 2024 – Aug 2024

- Designed the *CircuitLasso*, a scalable circuit learning method for mechanistic interpretability of LLMs, using sparse linear regression to efficiently uncover relationships among high-dimensional sparse-autoencoder features, revealing how interpretable semantic features propagate through the model, and improving downstream task performance. Published at *EMNLP 2026* (Main Conference).

IBM-Rensselaer AI Horizons Research Collaboration

Yorktown Heights, USA

Research Scientist Intern

May – Aug 2020, 2021, 2022

- Proposed an uncertainty-guided Bayesian inference framework to improve domain generalization performance in computer vision (CV) tasks. Introduced multi-level uncertainty quantification to evaluate model reliability. Published at *CVPR 2026*.
- Developed an invariant causal Markov blanket feature learning method to improve out-of-distribution generalization in CV tasks. Published at *ECCV 2024* and the *Causal Representation Learning Workshop, NeurIPS 2023*.
- Proposed an efficient causal discovery approach by proving the equivalence between the set of causal graphs and the set of weighted gradients of graph potential functions, hence enabling acyclicity-constraint-free causal graph learning over the equivalence space. Published at *ICPR 2024* (Oral) and *ICML 2021*.

Rensselaer Polytechnic Institute

Troy, USA

Graduate Research Assistant

May 2019 – Dec 2024, Full-time Ph.D. Student

- Built a unified framework combining causal discovery and representation learning for computer vision, designing structural-causal-model-based generative algorithms to capture and manipulate the image generation process; demonstrated counterfactual and controllable image generation via causal inference. (*Paper under review.*)
- Investigated causal discovery under heteroscedastic noise in structural equation models (SEMs); derived relaxed, implementable sufficient conditions proving identifiability for a general class of SEMs and proposed a two-phase iterative algorithm handling varying noise variance across variables and observations. Published at *AAAI 2024*.

AWARDS & HONORS

Travel grant, *Society for Industrial and Applied Mathematics – New York – New Jersey – Pennsylvania Section (SIAM-NNP)*, 2025 Annual Meeting.

Best Industrial Paper Award nomination, *ICPR 2024*.

PROFESSIONAL ACTIVITIES

Peer Review: Reviewer for NeurIPS, ICML, ICLR, CVPR, ICCV, AAAI, UAI, ECCV, AISTATS, and ARR, and for journals including the *Journal of Machine Learning Research (JMLR)*, *Journal of the American Statistical Association*, *IEEE Transactions on Information Theory*, *Computers & Mathematics with Applications* and *Communications Engineering*.

Conference Organization: Organized and hosted the minisymposium “Foundation Models: From Theory to Practice” at the *SIAM-NNP Annual Meeting*.